

Modular Arithmetic

Defⁿ: a, r and m (integers) we say that a is congruent to r modulo m and we write

$$a \equiv r \pmod{m} \Rightarrow m \mid (a-r)$$

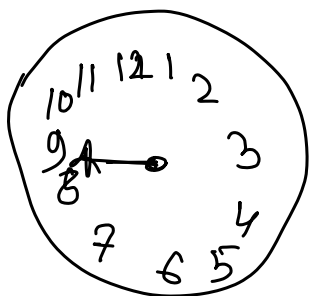
m divides $(a-r)$

example: $18 \equiv 4 \pmod{7} \Rightarrow 7 \mid (18-4)$

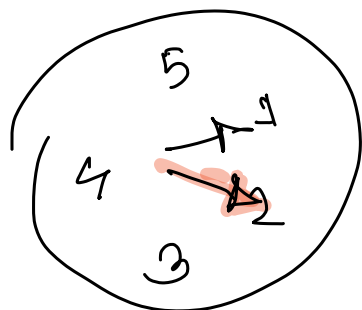
$$35 \equiv 0 \pmod{7} \Rightarrow 7 \mid (35-0)$$

$$-3 \equiv 1 \pmod{2} \Rightarrow 2 \mid (-3-1)$$

Clock metaphor



$$21 \equiv 9 \pmod{12}$$



$$17 \equiv 2 \pmod{5}$$

Proposition: Properties of modular

Assume, a, b, c, d and m some integers

$$a \equiv b \pmod{m} \text{ and } c \equiv d \pmod{m}$$

$$(i) \ a + c \equiv b + d \pmod{m}$$

$$(ii) \ a - c \equiv b - d \pmod{m}$$

$$(iii) \ a \cdot c \equiv b \cdot d \pmod{m}$$

Proof: Assume, a, b, c, d and m integers,

$$\text{and } \underline{a \equiv b \pmod{m}}, \quad \underline{c \equiv d \pmod{m}}$$

from
Defⁿ: $\downarrow$
 $\underline{m \mid (a-b)}$

from
Defⁿ: $\downarrow$
 $\underline{(a-b) = p \cdot m}$

$$\downarrow$$

 $\underline{m \mid (c-d)}$

$$\downarrow$$

 $\underline{(c-d) = q \cdot m}$

[p and q are integers]

$$a = p \cdot m + b \quad (i) \quad , \quad c = q \cdot m + d \quad (ii)$$

(i) $\cdot$ (ii)

$$a \cdot c = (p \cdot m + b) \cdot (q \cdot m + d)$$

$$\Rightarrow a \cdot c = \underline{pqm^2 + bqm + dpm} + bd$$

$$\Rightarrow a \cdot c = \underline{m} \left[\underline{(pqm + bq + dp)} \right] + \underline{bd}$$

integer

$$a \cdot c - b \cdot d = m \cdot (pqm + bq + dp)$$

$$\underline{m | (a \cdot c - b \cdot d)}$$

from defⁿ of modular arithmetic

$$a \cdot c = b \cdot d \pmod{m}$$

Defⁿ: (primes)

an integer $p \geq 2$ is prime if its only positive divisors are 1 and p (itself)

Lemma: a, b, c are integers, let p be a prime

(i) $p \nmid a$ then $\gcd(p, a) = 1$

idea: (sketch of proof)

divisors of p : 1, -1, p, -p

4 possible common divisors of a and p

1, -1, p, -p

- Which of the 4 divides both a and p
- Which one of the common divisors is the

largest?

→ key factor $p \nmid a$

Proof: Assume a, b, c integers and p : prime
and also, $p \nmid a$.

To be a common divisor of p and a
we need a number that divides both p and a

we know divisors of p : $1, -1, p, -p$,

also, p does not divide a ,

• p is not a common divisor of p and a .

• next largest choice of divisor = 1

we know, $1 \mid a$, and $1 \mid p$

1 is a common divisor

from the list of divisor 1 is the largest
after p ,

hence from defⁿ of gcd $\Rightarrow$ $\boxed{\gcd(a, p) = 1}$

[Proved]

(ii) a, b, c are integers, let p be a prime
if $a \mid bc$ and $\gcd(a, b) = 1$ then $a \mid c$

idea: we need a tool that connects integers with their $\gcd$ (Bezout's identity)

Proof: Assume $a \mid bc$ and $\gcd(a, b) = 1$

by Bezout's identity

there exists some integers k and l

$$\gcd(a, b) = a \cdot k + b \cdot l$$

since, $\gcd(a, b) = 1$ so,

$$1 = a \cdot k + b \cdot l$$

Now multiplying c on both side:

$$c = c a k + \underline{c b \cdot l}$$

Now, we know,

$a \mid bc$ by defⁿ, $bc = \boxed{a \cdot m}$ [$m: \text{int}$]

$$\therefore c = c a k + a m l$$

$$\therefore c = a (\underline{c k + m l}) \rightarrow \text{integer}$$

Defⁿ $\Rightarrow a|c$ (Proved)

(iii) if $p|bc$, then $p|b$ or $p|c$ (on both)

try

Proposition: Modular cancellation law:

let a, b, k and m are integers, if

$$ak \equiv bk \pmod{m} \text{ and } \gcd(m, k) = 1$$

$$\text{then } a \equiv b \pmod{m}$$

Proof: a, b, k and m : integers

$$\text{and also, } ak \equiv bk \pmod{m}$$

$$\text{and } \gcd(m, k) = 1$$

by defⁿ of modular arithmetic

$$m|(ak - bk)$$

by defⁿ of divisibility,

$$ak - bk = m \cdot l \quad [l \in \mathbb{Z}]$$

$$\rightarrow k(a - b) = m \cdot l$$

since $(a - b)$: integer,

$$k \mid m \cdot l$$

since we know $\gcd(k, m) = 1$

(ii) of previous lemma)

$$k \mid l$$

from defⁿ of divisibility, $l = k \cdot t \quad [t \in \mathbb{Z}]$

Now,

$$k(a-b) = m \cdot l \quad \text{~~not~~ } k(a-b) = m \cdot k \cdot t$$

$$\text{~~not~~ } (a-b) = m \cdot \underbrace{t}_{\text{integer}}$$

$$m \mid (a-b)$$

from defⁿ of modular Arithmetic

$$\boxed{a \equiv b \pmod{m}} \quad \text{~~not~~ [Proved]}$$

Theorem: (Fermat's little theorem)

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